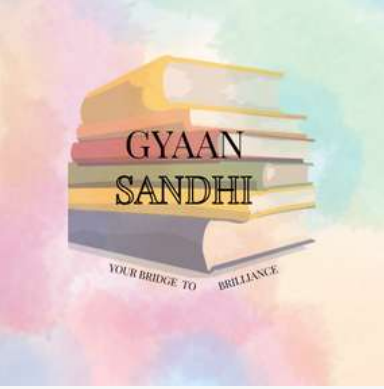




Information Security

Unit 4

**Data Integrity Algorithms
and Web Security**

Cryptographic Hash Functions

Definition

- A cryptographic hash function is a mathematical algorithm that converts data into a fixed-size hash value.
- It is used for security purposes like data integrity and password protection.

Characteristics

- Fixed length output
- Fast computation
- Deterministic (same input gives same output)
- One-way process
- Collision resistant

Cryptographic Hash Functions

Common Hash Algorithms

- MD5
- SHA-1
- SHA-256
- SHA-512

Applications

- Password storage
- Digital signatures
- Data integrity verification
- Blockchain technology
- File verification

Cryptographic Hash Functions

One-Way Hashing

Definition

- One-way hashing converts plain text into hash value which cannot be reversed back to original data.

Features

- Irreversible process
- Same input gives same hash
- Secure for storing passwords
- Used in authentication systems

Working Process

1. User enters data/password
2. Hash function generates hash value
3. Hash is stored in database
4. During login, input is hashed again and compared

Cryptographic Hash Functions

Advantages

- High security
- Protects passwords
- Fast verification

Disadvantages

- Original data cannot be recovered
- Collision attacks possible in weak algorithms

Examples

- SHA-256
- bcrypt
- Argon2

Two-Way Hashing (Encryption)

Cryptographic Hash Functions

Definition

- Two-way hashing actually refers to encryption where data can be encrypted and later decrypted using a key.

Features

- Reversible process
- Uses encryption and decryption keys
- Original data can be recovered

Types

Symmetric Encryption

- Same key for encryption and decryption
- Example: AES, DES

Asymmetric Encryption

- Public key and private key used
- Example: RSA

Cryptographic Hash Functions

Working Process

1. Plain text is encrypted
2. Cipher text is generated
3. Decryption key recovers original text

Advantages

- Data can be recovered
- Secure communication
- Protects confidential information

Disadvantages

- Key management required
- Slower than hashing

Applications

- Secure communication
- Online banking



Hash Functions Based on Cipher Block Chaining (CBC)

Introduction

- Cipher Block Chaining (CBC) is a mode of operation used in block ciphers.
- It can also be used to create cryptographic hash functions.
- In CBC-based hashing, each message block is processed one after another using encryption operations.

Basic Idea of CBC-Based Hash Function

- The message is divided into fixed-size blocks.
- Each block is combined with the previous encrypted output using XOR operation.
- The result is encrypted using a block cipher algorithm.
- The final encrypted block becomes the hash value.

Working of CBC-Based Hash Function

Hash Functions Based on Cipher Block Chaining (CBC)

Step-by-Step Process

Step 1: Divide Message into Blocks

- Message is divided into equal-sized blocks:

Example: M1, M2, M3, M4

Step 2: Initialize Initial Vector (IV)

- An initial value called IV is selected.
- Usually $IV = 0$ for hash functions.

Step 3: Process First Block

- First message block is XORed with IV.
- Result is encrypted.

Formula: $H1 = E(M1 \oplus IV)$

Where: E = Encryption function , $\oplus$ = XOR operation



Hash Functions Based on Cipher Block Chaining (CBC)

Step 4: Process Next Blocks

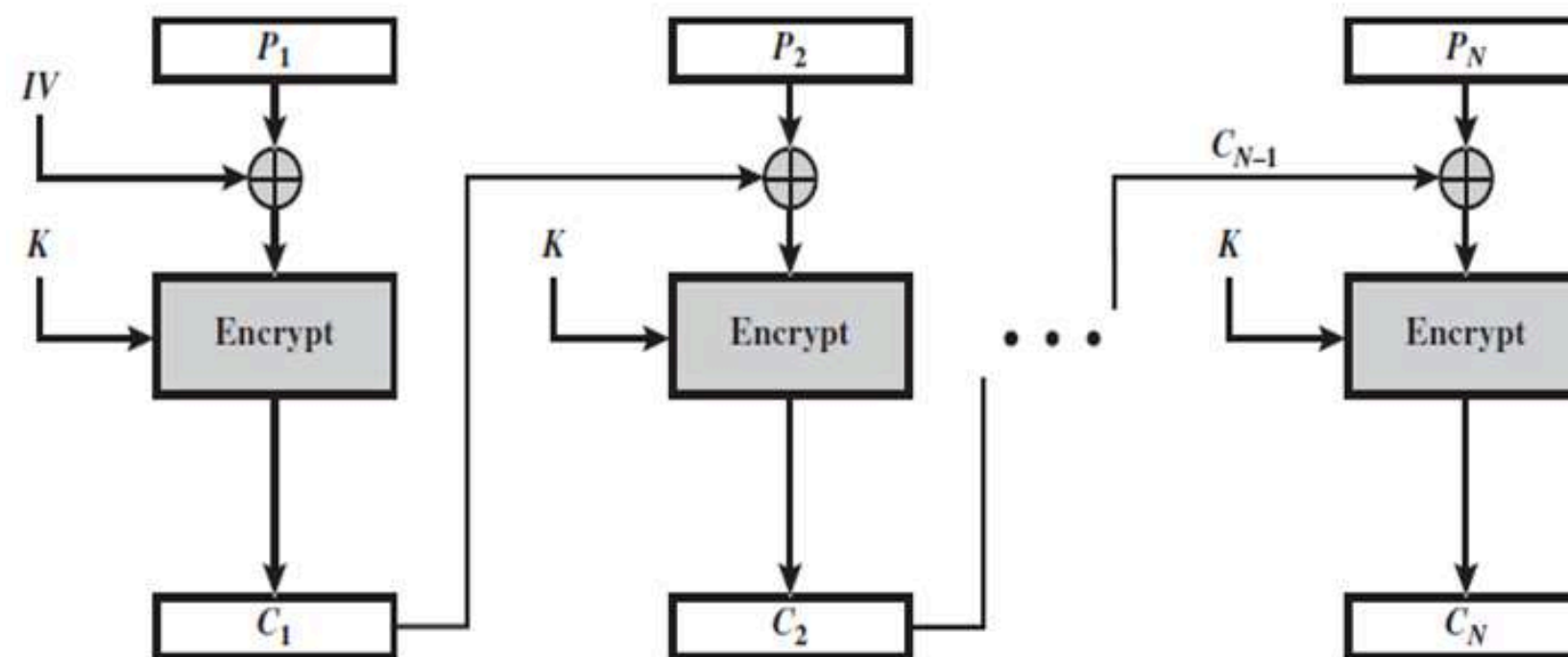
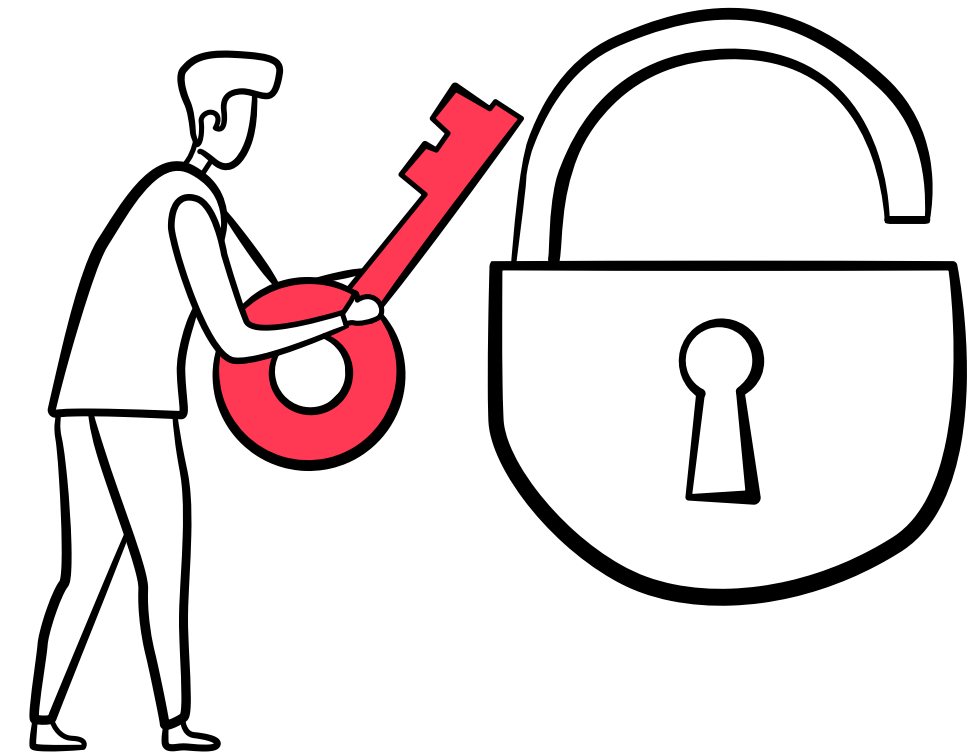
- Each next block is XORed with previous hash output.
- Then encrypted again.

Formula: $H2 = E(M2 \oplus H1)$, $H3 = E(M3 \oplus H2)$, $H4 = E(M4 \oplus H3)$

Step 5: Final Hash Value

- Last encrypted output becomes the final hash.

Hash = $H4$



Hash Functions Based on Cipher Block Chaining (CBC)

Characteristics of CBC-Based Hash Functions

1. Sequential Processing

- Each block depends on previous block output.

2. Avalanche Effect

- Small change in message changes entire hash output.

3. Fixed Length Output

- Output hash size remains fixed.

4. High Sensitivity

- Any modification in input changes final hash completely.

Advantages

- Provides good data integrity
- Difficult to predict final hash
- Uses strong encryption algorithms
- Secure against simple attacks



SHA-1

SHA-1 is a cryptographic hash function developed by the NSA.
It produces a 160-bit (20-byte) hash value.

FEATURES

- Produces 160-bit hash output
- One-way hash function
- Fixed length output
- Deterministic
- Fast computation
- Sensitive to input changes
- Based on Merkle–Damgård construction

HIGH LEVEL WORKING

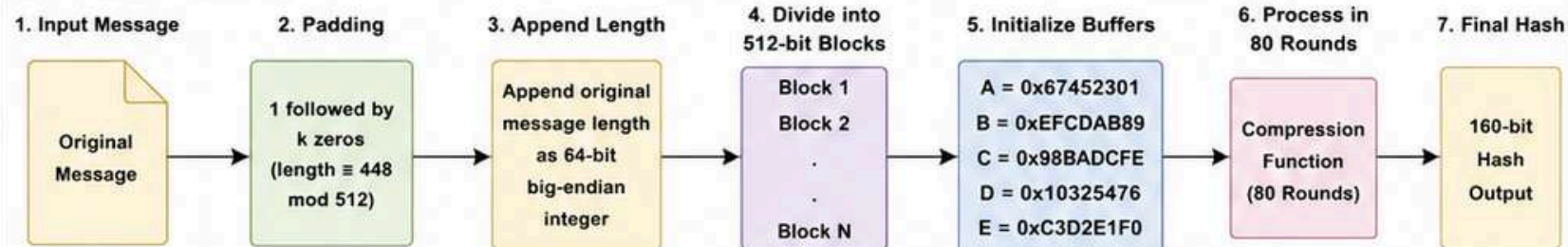
1. Input message
2. Padding
3. Append length
4. Divide into 512-bit blocks
5. Initialize five 32-bit buffers (A, B, C, D, E)
6. Process each block in 80 rounds
7. Produce final 160-bit hash

OUTPUT

160-bit (20-byte) Hash Value

Example:
Input: "Hello"
SHA-1:
f7ff9e8b7bb2b91af11f0b6a7b6d0f8b8b5d9e3f

WORKING OF SHA-1



COMPRESSION FUNCTION (80 ROUNDS)

- Each 512-bit block is processed in 80 rounds.
- In each round, a 32-bit word is generated using logical functions and left rotation.
- Uses four logical functions:

Rounds	Function	f(B,C,D)	Constant K
0 – 19	f1	(B AND C) OR ((NOT B) AND D)	0x5A827999
20 – 39	f2	B XOR C XOR D	0x6ED9EBA1
40 – 59	f3	(B AND C) OR (B AND D) OR (C AND D)	0x8F1BBCDC
60 – 79	f4	B XOR C XOR D	0xCA62C1D6

LOGICAL OPERATIONS USED

- AND, OR, XOR, NOT
- Left Circular Shift (rotate left)
- Addition modulo 2^{32}

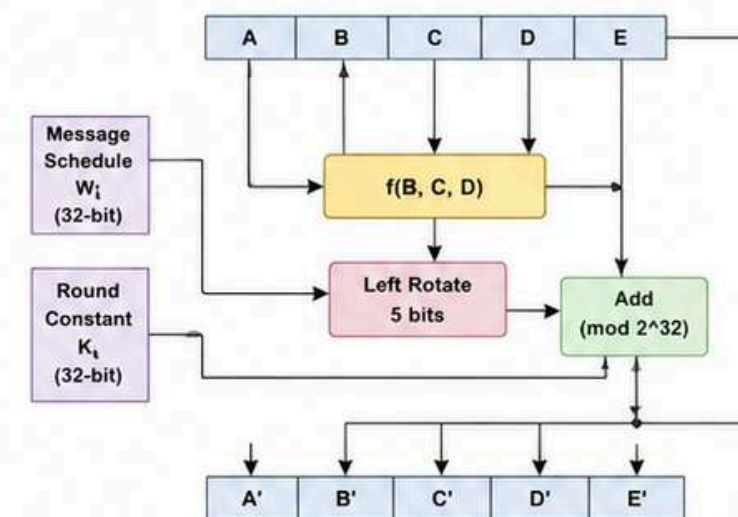


EXAMPLE

Input : Hello
SHA-1 : f7ff9e8b7bb2b91af11f0b6a7b6d0f8b8b5d9e3f

160 bits
20 bytes

INTERNAL ROUND OPERATION



ADVANTAGES

- ✓ Fast and efficient
- ✓ Easy to implement
- ✓ Widely used in past
- ✓ Good for integrity verification

DISADVANTAGES

- ✗ Vulnerable to collision attacks
- ✗ No longer considered secure
- ✗ Replaced by SHA-2 and SHA-3

APPLICATIONS



Data Integrity
Verification



Digital
Signatures



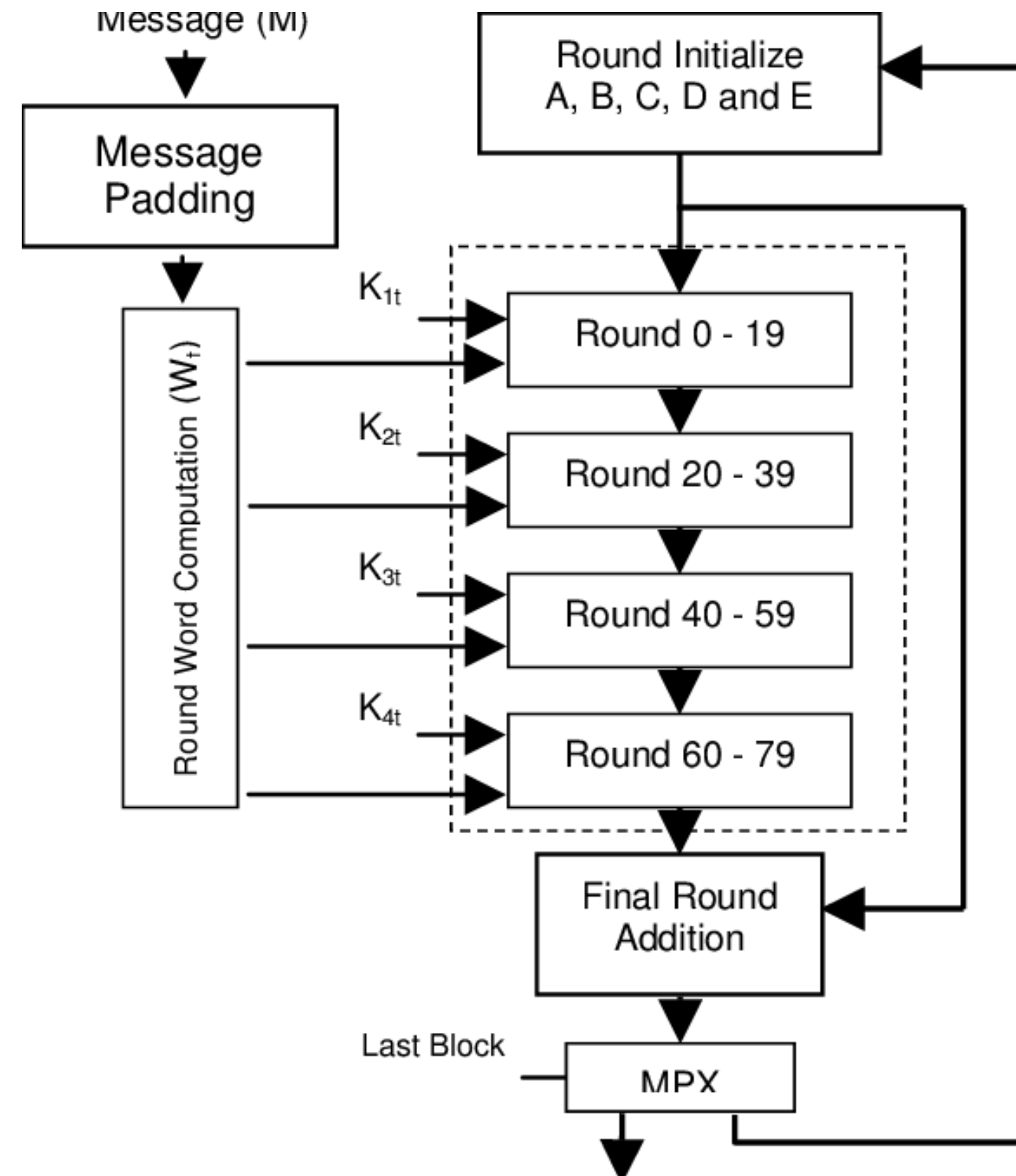
SSL Certificates
(earlier use)



File
Verification



Version Control
(e.g., Git)



Message Digest

Definition

- A message digest is a fixed-size numerical representation of data generated using a hash function.
- It is also called a hash value or digital fingerprint of the message.

Purpose of Message Digest

- Verify data integrity
- Detect changes in data
- Secure password storage
- Used in digital signatures
- Authentication and security



Message Digest

Examples of algorithms:

- MD5
- SHA-1
- SHA-256

Step 3: Generate Digest

- Fixed-size digest is produced.

Example:

a591a6d40bf420404a011733cfb7b190

Features of Message Digest

1. Fixed Output Size

- Output size remains constant.

2. Avalanche Effect



Message Digest

3. Irreversible

- Original message cannot be recovered.

4. Collision Resistant

- Difficult for two messages to have same digest.

Advantages

- Ensures data integrity
- Fast processing
- Secure authentication
- Easy verification

Disadvantages

- Weak algorithms can face collision attacks
- Cannot recover original data



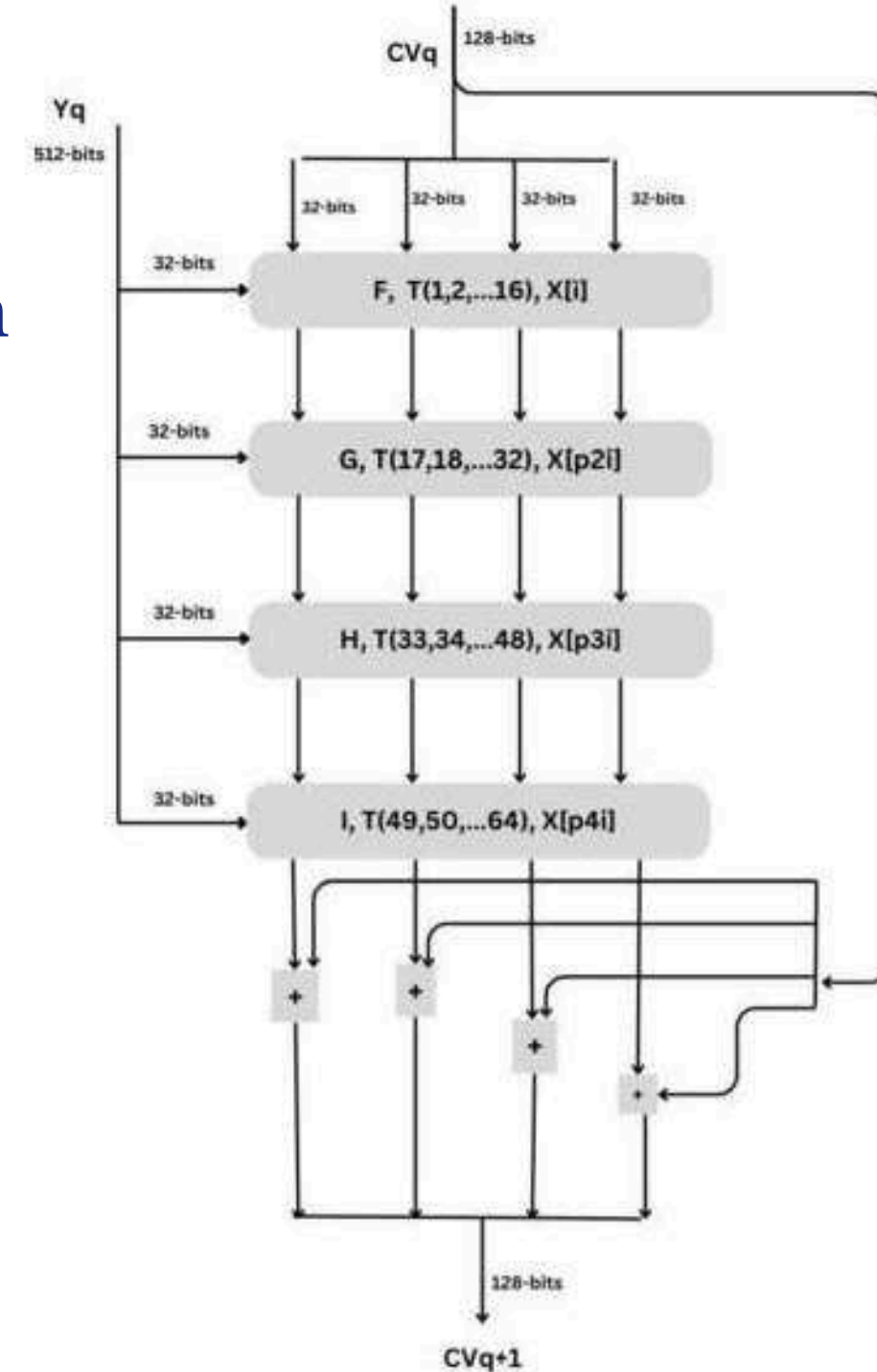
MD5

Introduction

- MD5 stands for Message Digest Algorithm 5.
- It is a cryptographic hash function developed by Ron Rivest in 1991.
- MD5 generates a 128-bit hash value.
- It is mainly used for data integrity verification.

Features of MD5

- Produces 128-bit hash output
- Fast hash generation
- One-way hashing function
- Fixed length output
- Deterministic algorithm
- Easy implementation



MD5

Working of MD5

Step 1: Input Message

- Original message/data is given as input.

Example: Hello World

Step 2: Padding

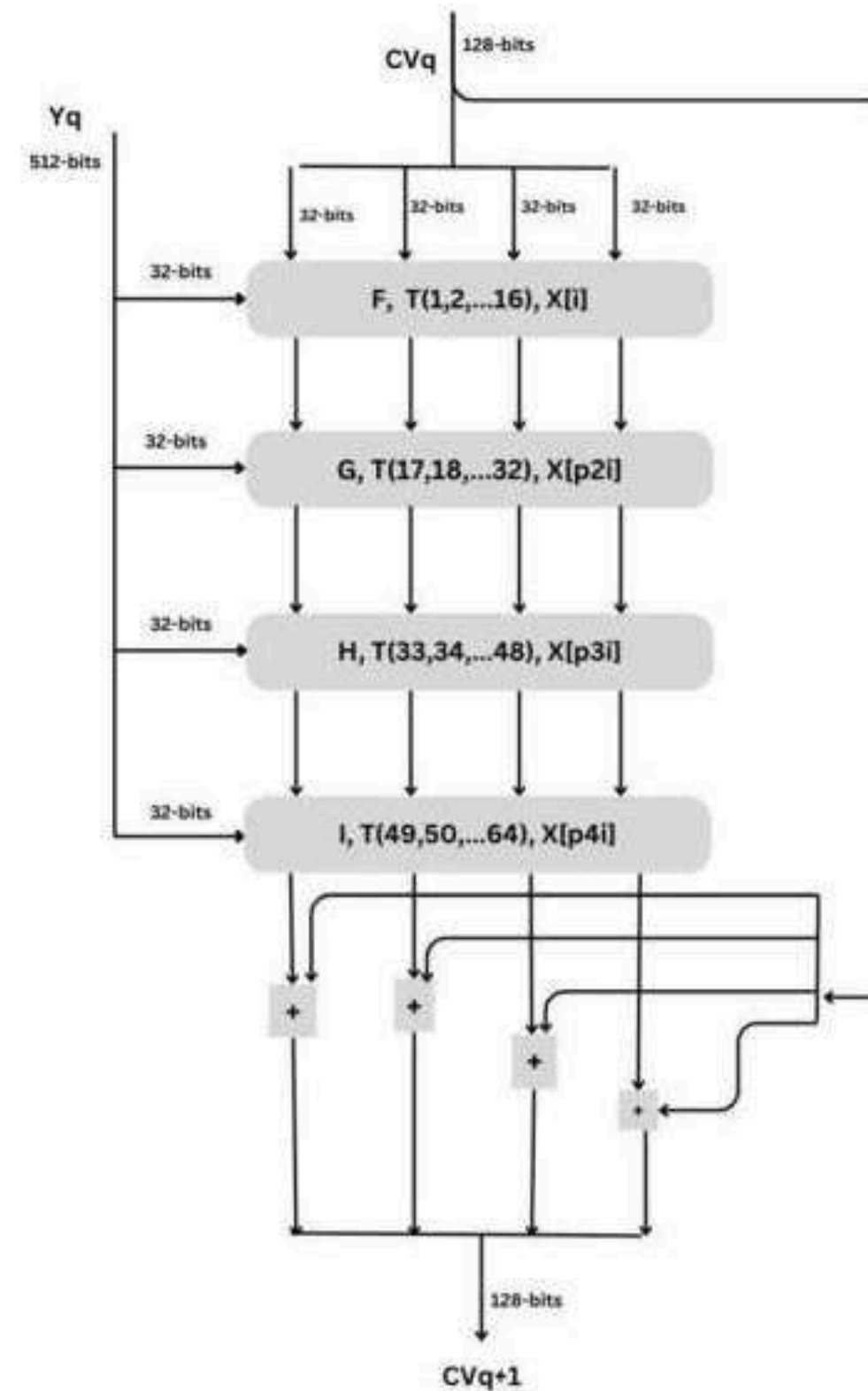
- Extra bits are added to the message.
- Message length becomes:

$\text{Length} \equiv 448 \pmod{512}$

- A 1 bit followed by 0 bits is added.

Step 3: Append Message Length

- Original message length is added as a 64-bit value.



MD5

Step 4: Divide into Blocks

- Message is divided into 512-bit blocks.

Example: Block 1, Block 2, Block 3 ...

Step 5: Initialize Buffers

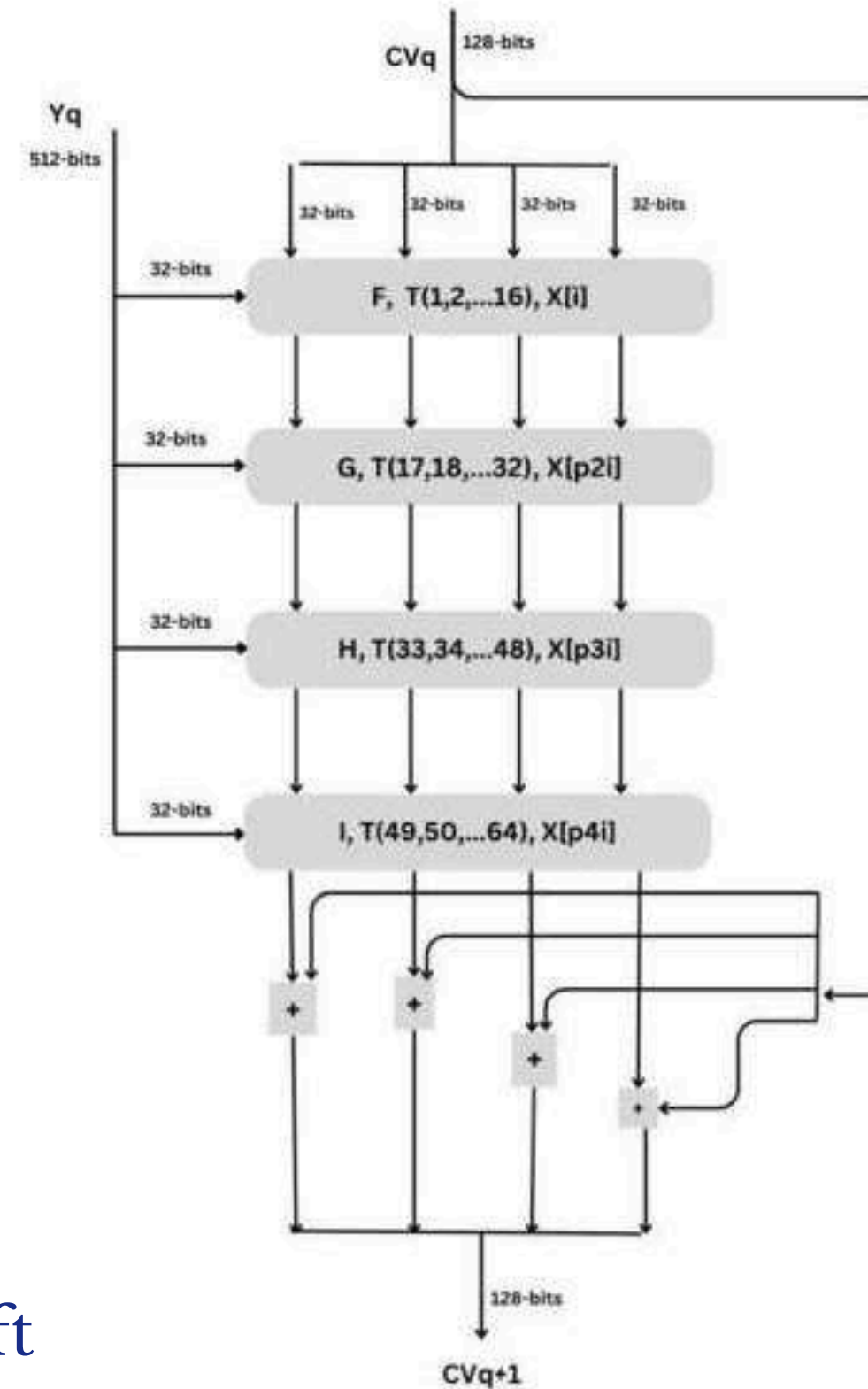
MD5 uses four 32-bit buffers: A, B, C, D

These buffers store intermediate results.

Step 6: Processing Rounds

- Each block goes through 4 rounds.
- Total of 64 operations are performed.

Operations include: AND , OR , XOR , NOT , Left Circular Shift



MD5

Step 4: Divide into Blocks

- Message is divided into 512-bit blocks.

Example: Block 1, Block 2, Block 3 ...

Step 5: Initialize Buffers

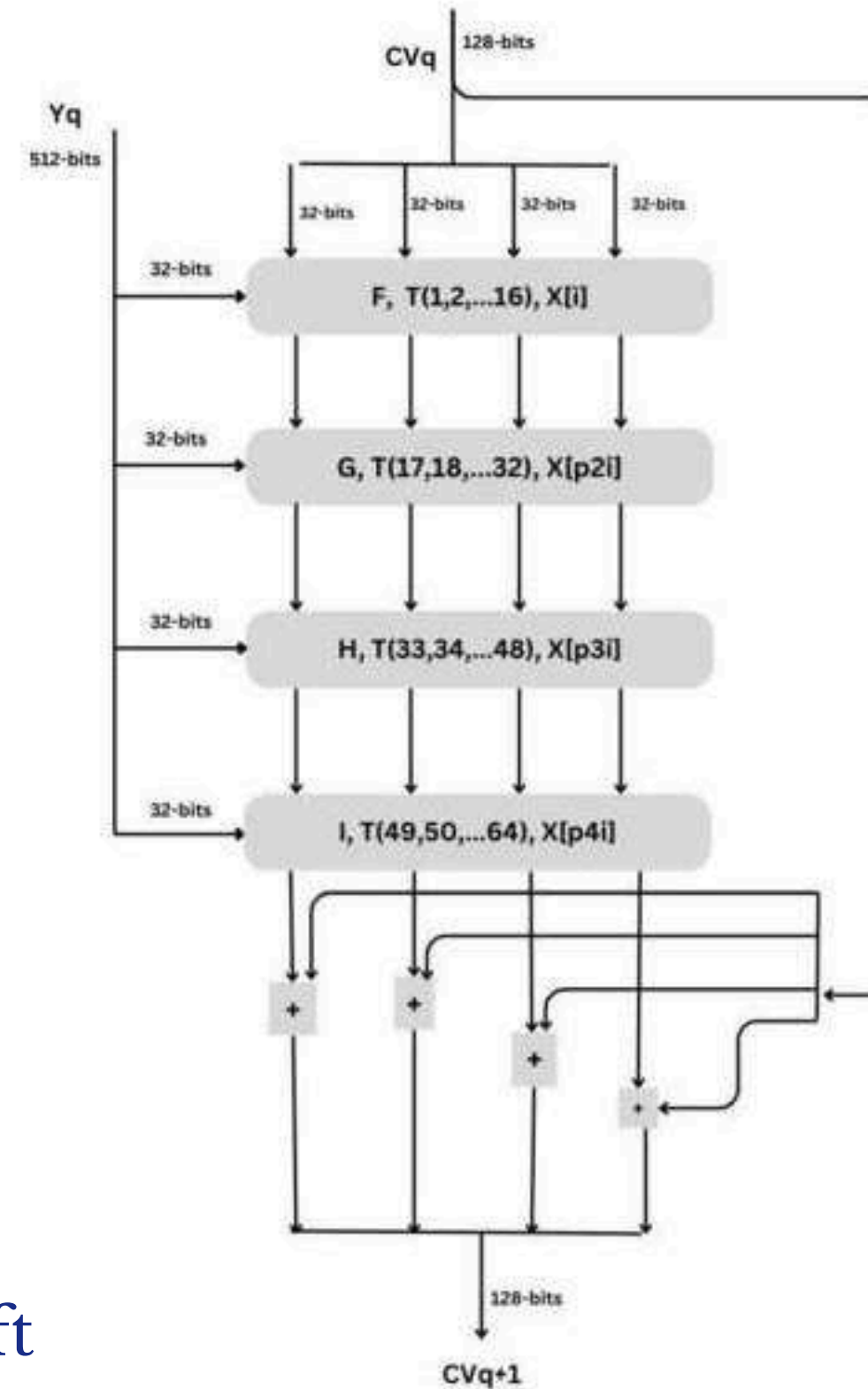
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- Total of 64 operations are performed.

Operations include: AND , OR , XOR , NOT , Left Circular Shift



Message Authentication Code (MAC)

Introduction

- Message Authentication Code (MAC) is a security technique used to verify:
 - Message authenticity
 - Message integrity
- It ensures that the message has not been modified during transmission.
- MAC uses a secret key shared between sender and receiver.

Definition

- A Message Authentication Code (MAC) is a fixed-size value generated from:
 - The original message
 - A secret key
- It is attached to the message before sending.

Message Authentication Code (MAC)

Objectives of MAC

- Verify the identity of sender
- Detect message modification
- Provide data integrity
- Protect against unauthorized access

Working of MAC

1. Sender takes the original message.
2. A secret key is applied using MAC algorithm.
3. MAC value (tag) is generated.
4. Message + MAC are sent to receiver.
5. Receiver uses same secret key to generate MAC again.

Message Authentication Code (MAC)

6. If both MAC values match:

- Message is authentic
- Message is unchanged

Formula of MAC :- $MAC = F(K, M)$

Where:

- K = Secret Key
- M = Message
- F = MAC Algorithm

Components of MAC

- Message
- Secret Key
- MAC Algorithm
- Authentication Tag

Example of MAC

- User sends:
- “Transfer ₹5000”
- MAC generated using secret key
- Receiver checks MAC
- If attacker changes amount to ₹9000:
- MAC mismatch occurs
- Message is rejected

Applications of MAC

- Online banking
- Secure emails
- Digital communication
- Network security protocols

MAC Based on DES (Data Encryption Standard)

Introduction

- DES-based MAC is a Message Authentication Code technique that uses the DES encryption algorithm.
- It is used to provide:
 - Message integrity
 - Message authentication
- The sender and receiver share the same secret key.

What is DES?

- DES (Data Encryption Standard) is a symmetric key block cipher.
- It encrypts data in:
 - 64-bit blocks
- Uses:
 - 56-bit secret key
- Developed for secure communication.

MAC Based on DES (Data Encryption Standard)

DES-Based MAC Concept

- In DES-based MAC:
 - Message is divided into fixed-size blocks.
 - DES encryption is applied repeatedly.
 - Final encrypted block becomes the MAC value.

Working of DES-Based MAC

Step 1: Divide Message

- Original message is divided into 64-bit blocks:
 - $M_1, M_2, M_3, \dots, M_n$

Step 2: Apply DES Encryption

- First block is encrypted using secret key K .

$$[C_1 = E_K(M_1)]$$

MAC Based on DES (Data Encryption Standard)

Step 3: XOR Operation

- Next message block is XORed with previous encrypted output.

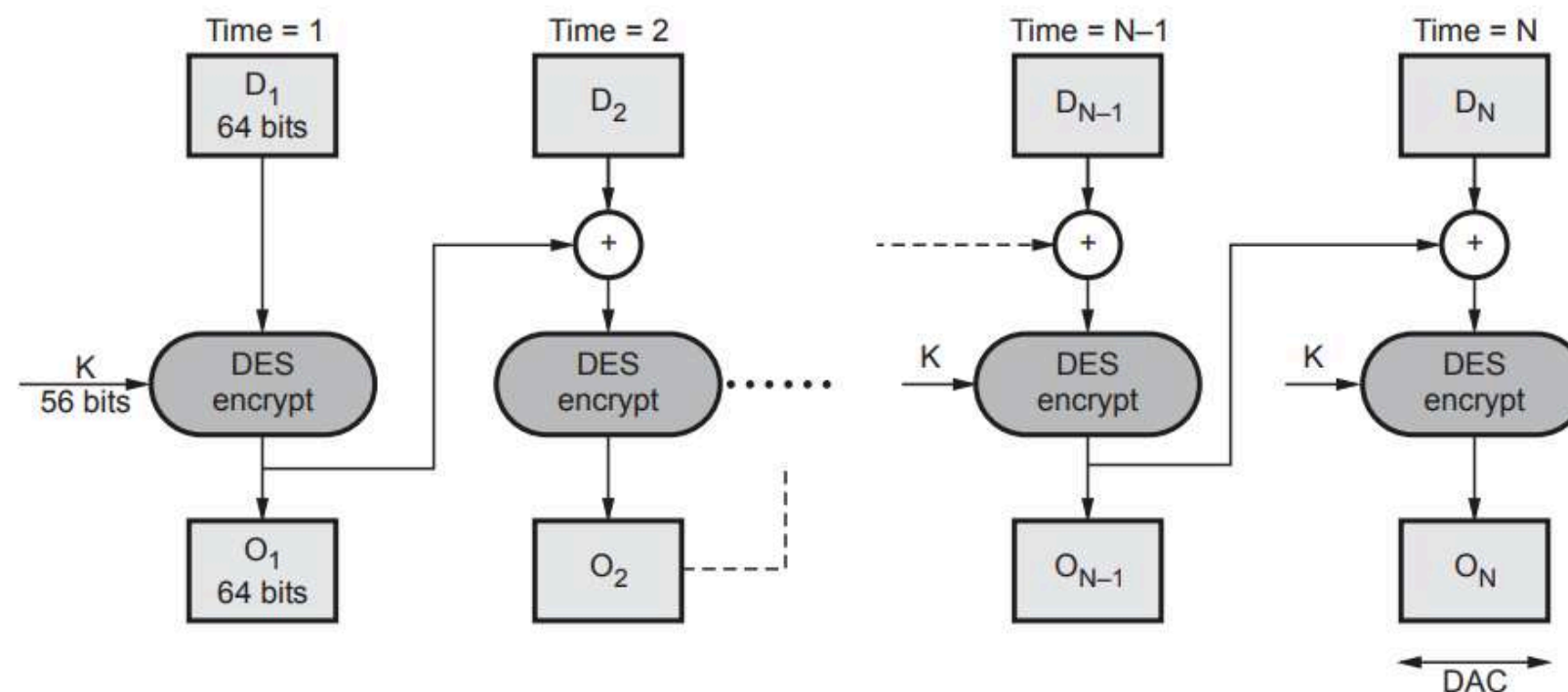
$$[C_2 = E_K(M_2 \oplus C_1)]$$

- Process continues for all blocks.

Step 4: Final Output

- Last encrypted block becomes the MAC.

$$[MAC = C_n]$$



MAC Based on DES (Data Encryption Standard)

Features of DES-Based MAC

- Uses symmetric key cryptography
- Detects message modification
- Ensures sender authenticity
- Fixed-length MAC output

Advantages

- Simple implementation
- Fast authentication process
- Good integrity checking
- Suitable for secure communication

Disadvantages

- DES is now considered weak
- Vulnerable to brute-force attacks
- Less secure compared to AES-based MAC

Applications

- Banking systems
- Secure transactions
- Network authentication
- Legacy security systems

Digital Signature, DSS and DSS Algorithm

Introduction to Digital Signature

- A Digital Signature is a cryptographic technique used to verify:
 - Authenticity of sender
 - Integrity of message
 - Non-repudiation
- It acts like an electronic signature for digital documents.

Definition

- A Digital Signature is created using:
 - Sender's private key
- It is verified using:
 - Sender's public key

Digital Signature, DSS and DSS Algorithm

Objectives of Digital Signature

- Verify sender identity
- Ensure message integrity
- Prevent message tampering
- Provide non-repudiation

Working of Digital Signature

Step 1: Message Creation

- Sender creates the original message.

Step 2: Hash Generation

- Hash function generates message digest. $[\text{Hash} = H(M)]$

Where:

- H = Hash Function
- M = Message

Digital Signature, DSS and DSS Algorithm

Step 3: Signature Creation

- Message digest is encrypted using sender's private key.

$$[DS = E_{\{PR\}}(Hash)]$$

Where:

- DS = Digital Signature
- PR = Private Key

Step 4: Sending Data

- Sender sends:
 - Original message
 - Digital Signature

Step 5: Verification

- Receiver decrypts signature using sender's public key.

Digital Signature, DSS and DSS Algorithm

DSS (Digital Signature Standard)

Introduction

- DSS stands for Digital Signature Standard.
- Developed by:
 - NIST (National Institute of Standards and Technology)
- Used for creating and verifying digital signatures.

Features of DSS

- Uses public key cryptography
- Based on DSA algorithm
- Provides authentication and integrity
- Does not provide encryption

Digital Signature, DSS and DSS Algorithm

Components of DSS

- DSA (Digital Signature Algorithm)
- SHA (Secure Hash Algorithm)
- Public Key
- Private Key

DSA (Digital Signature Algorithm)

Definition

- DSA is the algorithm used in DSS for generating digital signatures.

Working of DSA Algorithm

Step 1: Key Generation

- Generate: Private Key (x)
 - Public Key (y)

Digital Signature, DSS and DSS Algorithm

Step 2: Signature Generation

- Create hash of message.
- Use private key to generate signature values:
 - r
 - s

Signature: $[r,s]$

Step 3: Send Message

- Sender sends:
 - Original message
 - Signature (r,s)

Step 4: Signature Verification

- Receiver:
 - Computes hash value
 - Uses sender public key
 - Verifies signature
- If verified:
 - Message is authentic

Digital Signature, DSS and DSS Algorithm

DSS (DSA) ALGORITHM STEPS

1. SIGNATURE GENERATION

1 Select random number k

Choose a random integer k such that $1 < k < q$

2 Compute r

$$r = (g^k \bmod p) \bmod q$$

If $r = 0$, choose another k

3 Compute s

$$s = k^{-1}(H(M) + xr) \bmod q$$

If $s = 0$, choose another k

4 Signature

$$\text{Signature} = (r, s)$$

Send the message M along with signature (r, s)

Where,

- p – large prime number
- q – prime number, divisor of $(p - 1)$
- g – generator, $1 < g < p$
- x – private key, $0 < x < q$
- y – public key, $y = g^x \bmod p$
- $H(M)$ – hash of message M

2. SIGNATURE VERIFICATION

1 Compute w

$$w = s^{-1} \bmod q$$

(s^{-1} is the modular multiplicative inverse of $s \bmod q$)

2 Compute u_1 and u_2

$$u_1 = H(M)w \bmod q$$

$$u_2 = rw \bmod q$$

3 Compute v

$$v = ((g^{u_1} y^{u_2}) \bmod p) \bmod q$$

4 Verification

If $v = r$
Then signature is valid.

Otherwise, signature is invalid.

Where,

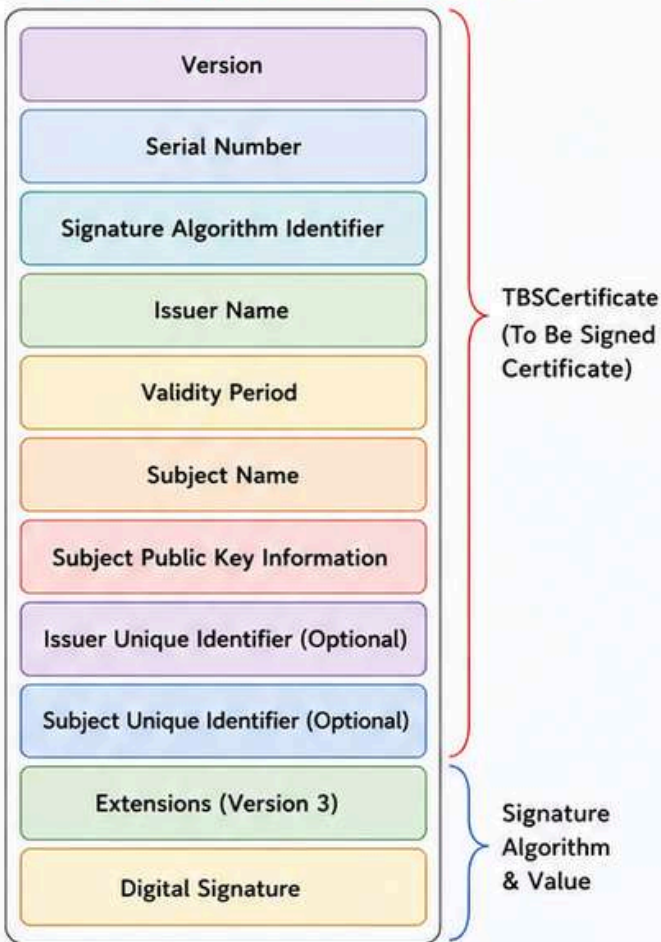
- M – original message
- (r, s) – signature received
- y – public key of the signer
- $H(M)$ – hash of message M using the same hash function

X.509 Certificate Format

X.509 is a standard format for Public Key Certificates used in secure communication and digital signatures.

Field	Description	Example
1 Version	Version of X.509 certificate being used.	V3
2 Serial Number	Unique serial number assigned by CA.	01:23:45:67:89:AB
3 Signature Algorithm Identifier	Algorithm used by CA to sign the certificate.	sha256WithRSA Encryption
4 Issuer Name	Name of the Certificate Authority (CA).	CN = DigiCert Inc, O = DigiCert Inc, C = US
5 Validity Period	Period for which the certificate is valid.	Not Before: 01-Jan-2024 Not After: 01-Jan-2025
6 Subject Name	Name of the certificate owner (subject).	CN = www.example.com, O = Example Ltd, C = IN
7 Subject Public Key Information	Contains the public key and the algorithm information.	Algorithm: RSA Public Key: (2048-bit)
8 Issuer Unique Identifier (Optional)	Unique identifier for the issuer.	(Optional)
9 Subject Unique Identifier (Optional)	Unique identifier for the subject.	(Optional)
10 Extensions (Version 3)	Additional information like Key Usage, SAN, Policies, etc.	Key Usage Basic Constraints Subject Alt Name
11 Digital Signature	Digital signature of CA on the certificate.	(Signature Value)

X.509 CERTIFICATE STRUCTURE



WORKING OF X.509 CERTIFICATE



FEATURES

- Standard format for public key certificates
- Supports Public Key Infrastructure (PKI)
- Provides authentication and integrity
- Used in SSL/TLS, email security, VPN, etc.



APPLICATIONS

- HTTPS websites
- SSL/TLS security
- VPN authentication
- Secure emails
- Digital signatures

TLS (Transport layer Security)

- Transport Layer Security (TLS) is a security protocol used to provide secure communication over a network.
- It is the successor of SSL (Secure Socket Layer).

TLS ensures:

- Confidentiality using encryption
- Integrity using hashing/MAC
- Authentication using digital certificates

TLS is commonly used in:

- Web browsing (HTTPS)
- Online banking
- Email security
- E-commerce websites

TLS (Transport layer Security)

Services Provided by TLS

1. Confidentiality

- Data is encrypted so unauthorized users cannot read it.

2. Integrity

- Ensures data is not modified during transmission.

3. Authentication

- Verifies the identity of the server and optionally the client.

TLS Handshake Protocol

- TLS Handshake Protocol is used to establish a secure session between client and server.
- During handshake:
 - Client and server authenticate each other
 - Encryption algorithms are selected
 - Secret keys are generated

Steps in TLS Handshake

1. Client Hello

TLS (Transport layer Security)

- Client sends:
 - TLS version
 - Supported cipher suites
 - Random number

2. Server Hello

- Server responds with:
 - Selected cipher suite
 - Server random number
 - Digital certificate

3. Server Authentication

- Client verifies the server certificate using Certificate Authority (CA).

4. Key Exchange

- Client and server exchange information to generate a shared secret key.

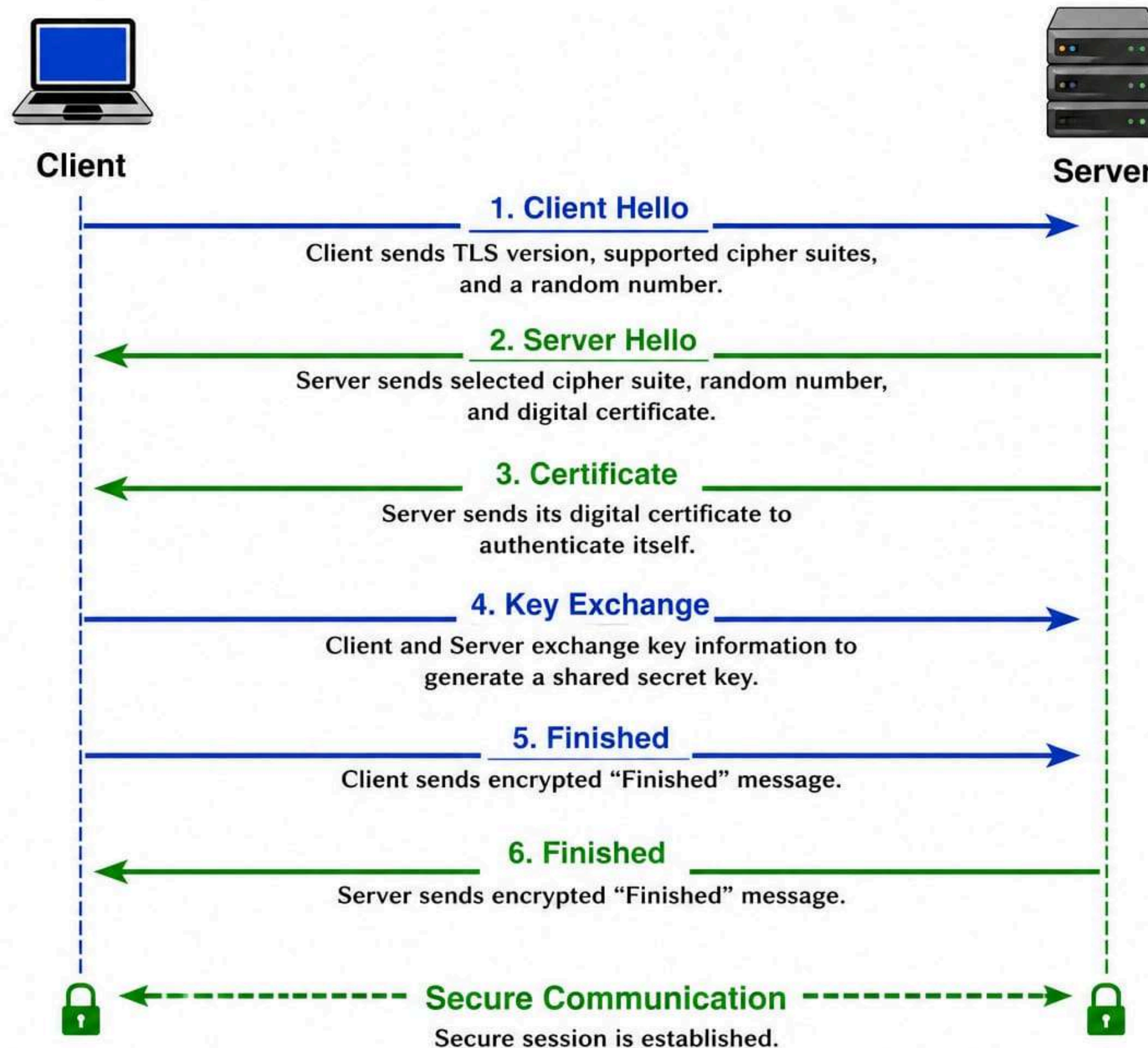
5. Session Key Generation

- Both generate the same session key independently.

6. Finished Messages

- Client and server send encrypted “Finished” messages.
- Secure communication starts after this step.

TLS (Transport Layer Security)



Comparison between IPSEC and TLS

IPSec	TLS
Works at Network Layer	Works at Transport Layer
Secures entire IP packets	Secures application data
Used mainly in VPNs	Used mainly in HTTPS and web security
Protects all applications automatically	Protects only specific applications
More complex configuration	Easier to configure
Uses AH and ESP protocols	Uses SSL/TLS protocols
Can work in Tunnel Mode and Transport Mode	Works only through secure sessions
Provides network-to-network security	Provides end-to-end communication security
Suitable for site-to-site communication	Suitable for browser-to-server communication
Commonly used in corporate networks	Commonly used in web applications

Email Security

- Email security protects email communication from unauthorized access and cyber attacks.
- It ensures confidentiality, integrity, and authenticity of emails.
- Email security helps prevent spam, phishing, malware, and email spoofing.
- Encryption is used to convert emails into unreadable form for protection.
- Digital signatures verify sender identity and message integrity.
- Spam filters block unwanted and harmful emails.
- Antivirus software scans email attachments for viruses and malware.
- PGP and S/MIME are commonly used email security technologies.
- Email authentication methods include passwords, OTP, and multi-factor authentication.
- Email security is widely used in banking, businesses, government, and online services.

IPv4 Header Format

0	3	4	7	8	15	16	18	19	31
VER 4 bits				HEL 4 bits		Service type 8 bits		Total length 16 bits	
Datagram identification 16 bits								Flags 3 bits	Fragment offset 13 bits
Time to live 8 bits						Protocol 8 bits		Header checksum 16 bits	
Source IP address 32 bits									
Destination IP address 32 bits									
Options									

Fig. 4.12.1 IPv4 header format

- In the IPv4 layer, packets are called datagrams.

An IPv4 datagram consists of two main parts:

- Header
- Data (Payload)

IPv4 Header Format

- The header contains control information needed for routing and delivery of packets.
- The IPv4 header has a variable length.
- HLEN (Header Length) Field
- HLEN indicates the length of the IPv4 header.
- The length is measured in 32-bit words.

VER (Version) Field

- The VER field specifies the version of the Internet Protocol being used.
- Current IPv4 version number is 4.
- Version 5 was used for experimental purposes.
- Version 6 represents IPv6, the next generation Internet Protocol.
- Minimum header length = 5 words = 20 bytes.
- Maximum header length = 15 words = 60 bytes.
- It helps the receiver identify where the data section begins.

IPv4 Header Format

Service Type

- Indicates the Quality of Service (QoS) requested for the IP datagram.
- Specifies how the packet should be handled during transmission.
- The last bit is reserved for future use.

Total Length

- Specifies the total size of the datagram.
- Includes both header and data length.
- Measured in octets (bytes).

Precedence	Types of service	R
------------	------------------	---

Precedence specifies the nature / priority :

000	Routine
001	Priority
010	Immediate
011	Flash
100	Flash override
101	Critical
110	Internetwork control
111	Internetwork control

TOS specifies the type of service value :

TOS bits	Description
1000	Minimize delay
0100	Maximum throughput
0010	Maximize reliability
0001	Minimize monetary cose
0000	Normal service

IPv4 Header Format

Identification

- A unique number assigned by the sender.
- Used during fragmentation and reassembly of datagrams.

Flags

- Contains control information for fragmentation.
- First bit: Reserved (must be 0).
- Second bit (DF – Do Not Fragment):
 - 0 = Fragmentation allowed
 - 1 = Fragmentation not allowed
- Third bit (MF – More Fragments):
 - 0 = Last fragment
 - 1 = More fragments follow

IPv4 Header Format

Fragment Offset

- Helps in reassembling fragmented datagrams.
- Indicates the position of the fragment in the original datagram.
- First fragment has offset value 0.

TTL (Time To Live)

- Specifies the lifetime of the datagram.
- Prevents packets from circulating endlessly in the network.
- Decreased by each router hop.

Protocol Number

- Identifies the higher-level protocol.
- Examples:
 - ICMP = 1
 - TCP = 6
 - UDP = 17

Header Checksum

- Used for error checking of the header.
- If checksum is incorrect, the datagram is discarded.

Source and Destination IP Address

- Contains sender and receiver IP addresses.

IPv4 Header Format

IP Options

- Variable-length field used for control, debugging, and testing.
- Examples:
 - Source routing
 - Timestamp recording

Padding

- Added to make the header length a multiple of 32 bits.
- Padding bits are set to zero.

IP Security (IPSec)

- IPSec is a security feature added to IPv4 and IPv6 for secure communication over LAN, WAN, and the Internet.
- It provides:
 - Confidentiality
 - Authentication
 - Message Integrity
 - Replay Detection
- IPSec works at the IP layer.
- The device implementing IPSec is called a Security Gateway (firewall or router).

Modes of IPSec

- Transport Mode Protects only the data part of the packet.
- Tunnel Mode Protects the entire IP packet by encapsulating it inside another packet.

IP Security (IPSec)

IPSec Protocols

- Authentication Header (AH) Provides authentication and integrity.
- Encapsulating Security Payload (ESP) Provides confidentiality, authentication, and integrity.

IP Security Architecture

- IPSec uses a Security Policy Database (SPD).
- SPD defines how packets should be handled and which security services are required.

Main Components of IPSec Architecture

- IPSec Documents
- IPSec Services
- Security Associations (SA)

IP Security (IPSec)

IPSec Documents

- IPSec specifications are defined through various standard documents.
- These documents describe:
 - Security protocols
 - Algorithms
 - Key management methods
 - Packet formats
- They help in implementing IPSec in a standard and secure way.

IPSec Services

IPSec provides the following security services:

- Authentication – Verifies the identity of sender and receiver.
- Confidentiality – Encrypts data to keep it secret.

IP Security (IPSec)

- Integrity – Ensures data is not modified during transmission.
- Replay Protection – Prevents attackers from re-sending old packets.
- Access Control – Allows only authorized users to access data.

Security Associations (SA)

- A Security Association (SA) is a logical connection between two devices using IPSec.
- It defines the security parameters for communication.
- SA includes:
 - Encryption algorithm
 - Authentication method
 - Secret keys
 - Security protocol used (AH or ESP)
- Each SA is identified by a unique Security Parameter Index (SPI).

Transport Mode

- In Transport Mode, only the data (payload) part of the IP packet is encrypted and protected.
- The original IP header remains unchanged.
- Mainly used for host-to-host communication.
- Provides security with less overhead and faster performance.

Working

- Original IP header is kept visible.
- IPSec header (AH or ESP) is added between IP header and data.
- Example
 - A laptop sending secure data directly to a server using IPSec.

Example: Communication between two computers in an office network.

Tunnel Mode

- In Tunnel Mode, the entire original IP packet (header + data) is encrypted.
- A new IP header is added outside the encrypted packet.
- Mainly used for gateway-to-gateway or VPN communication.
- Provides higher security than transport mode.

Working

- Entire original packet is encapsulated inside a new packet.
- New IP header is used for routing.

Example

- Secure communication between two company branch offices through VPN.
- Example: Office firewall to office firewall communication.

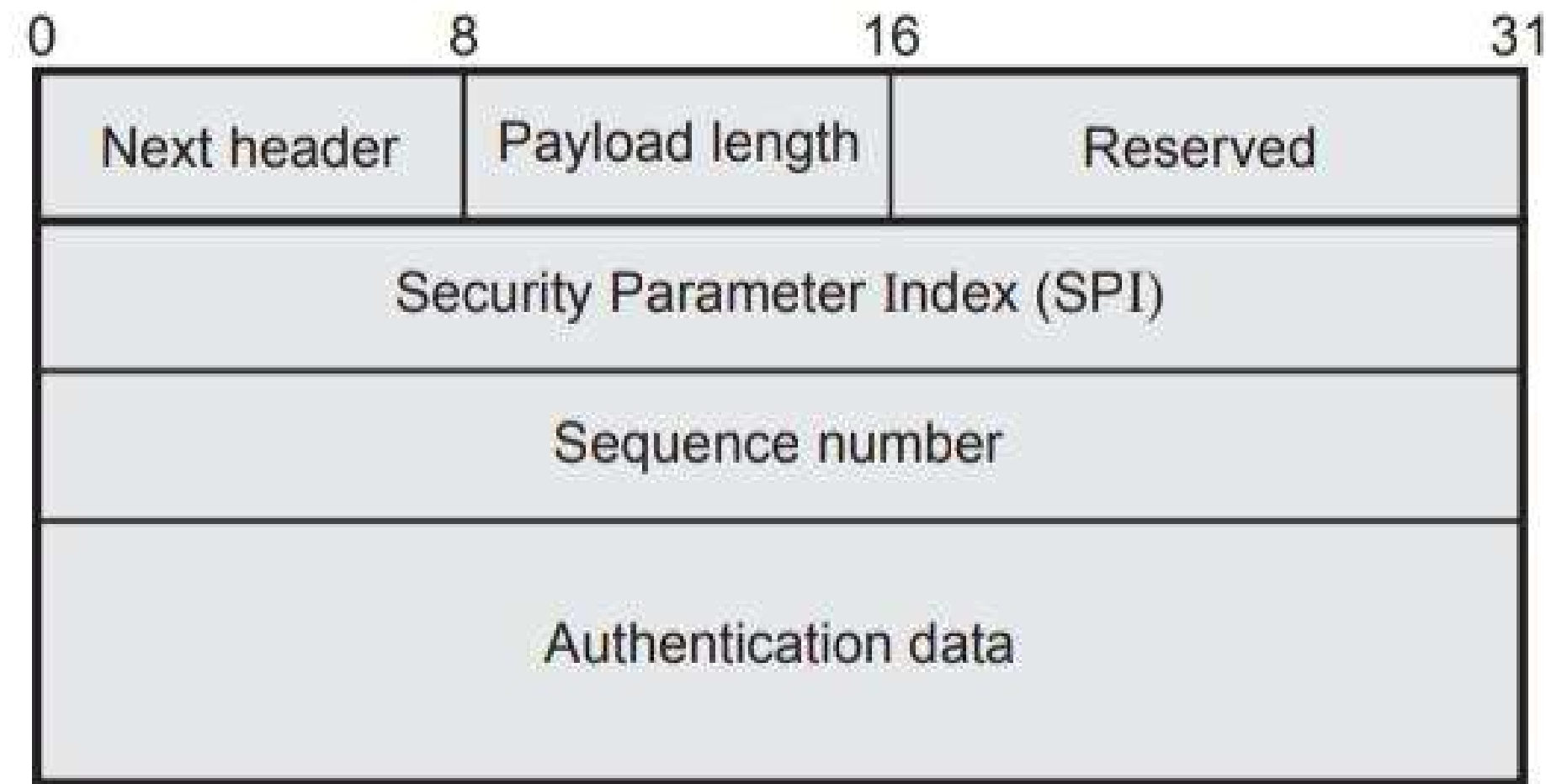
Advantages

- Hides original IP addresses
- More secure
- Commonly used in VPNs

Difference Between Transport Mode and Tunnel Mode

Transport Mode	Tunnel Mode
Only the data (payload) part of the IP packet is encrypted and protected.	The entire IP packet including header and data is encrypted and protected.
The original IP header remains visible during transmission.	The original IP header is hidden and a new IP header is added.
Mainly used for host-to-host communication.	Mainly used for gateway-to-gateway or VPN communication.
Provides lower security because header information is visible.	Provides higher security because the complete packet is hidden.
Packet size increase is smaller, so overhead is low and transmission is faster.	Packet size increases more due to encapsulation, resulting in higher overhead.
Commonly used in direct communication between two computers or devices.	Commonly used in Virtual Private Networks (VPNs) between two networks or security gateways.

Authentication Header



- Authentication Header (AH) provides data integrity, authentication, and protection against replay attacks for IP packets.
- It ensures that the data inside the IP packet is not modified during transmission.

Authentication Header

- AH verifies the identity of the sender and helps the receiver decide whether to accept or reject the packet.
- It prevents IP spoofing attacks.
- AH works using the MAC (Message Authentication Code) protocol.
- Both communicating parties must share a secret key.

AH Header Fields

1. Next Header

- 8-bit field.
- Identifies the type of header that follows the AH.

2. Payload Length

- Specifies the length of AH in 32-bit words minus 2.
- Used to determine the total size of the AH.

Authentication Header

3. Reserved

- 16-bit field reserved for future use.

4. Security Parameter Index (SPI)

- Identifies the Security Association (SA) for the communication.
- Used together with source/destination address and IPSec protocol.

5. Sequence Number

- Used to prevent Replay Attacks.
- Each packet is assigned a unique sequence number.

Replay Attack

- In a replay attack, an attacker captures a valid message and retransmits it.
- Example:

Authentication Header

- a. User A sends a fund transfer request to Bank B.
- b. Attacker/User C captures the message.
- c. The same message is resent to Bank B.
- d. Bank B processes it again unknowingly.
- e. Funds may be transferred twice.

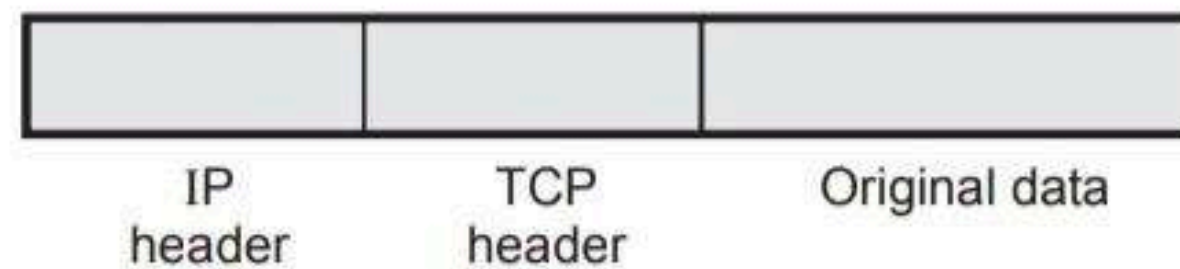
Authentication Data

- Also called Integrity Check Value (ICV).
- Contains the MAC value used for:
 - Authentication
 - Data integrity verification

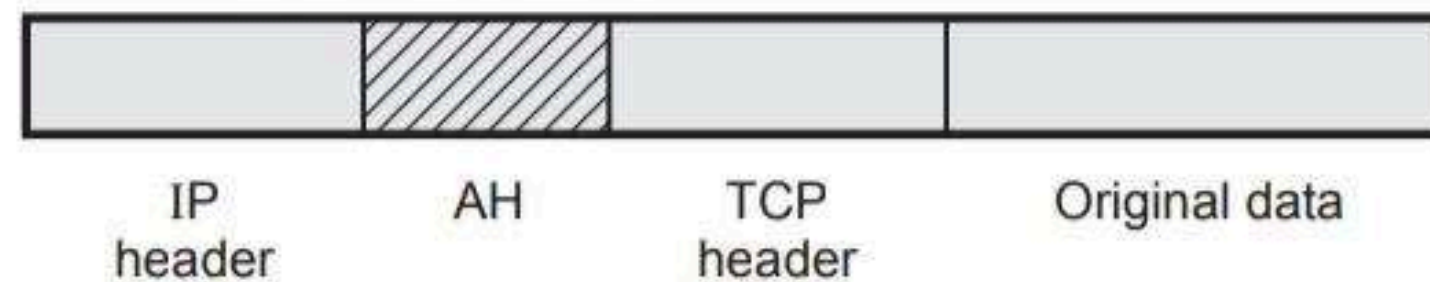
AH Transport mode

1. AH Transport Mode

- a. The position of the AH is between the original IP header and original TCP header of the IP packet.



(a) Before applying AH

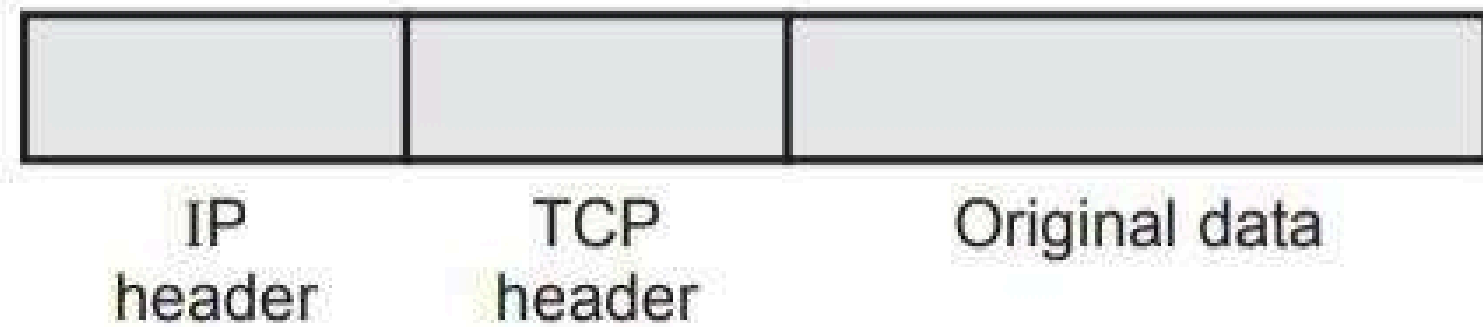


(b) After applying AH

AH Tunnel mode

AH Tunnel Mode

1. The entire original IP packet is authenticated.
2. AH is inserted between the original IP header and a new outer IP header.



(a) Before applying AH



(b) After applying AH

Fig. 4.14.3 Tunnel mode

ESP Format

- Encapsulating Security Payload (ESP) provides confidentiality by encrypting data.
- It also provides limited traffic flow confidentiality.
- Authentication service in ESP is optional.
- ESP protects the payload data from unauthorized access.

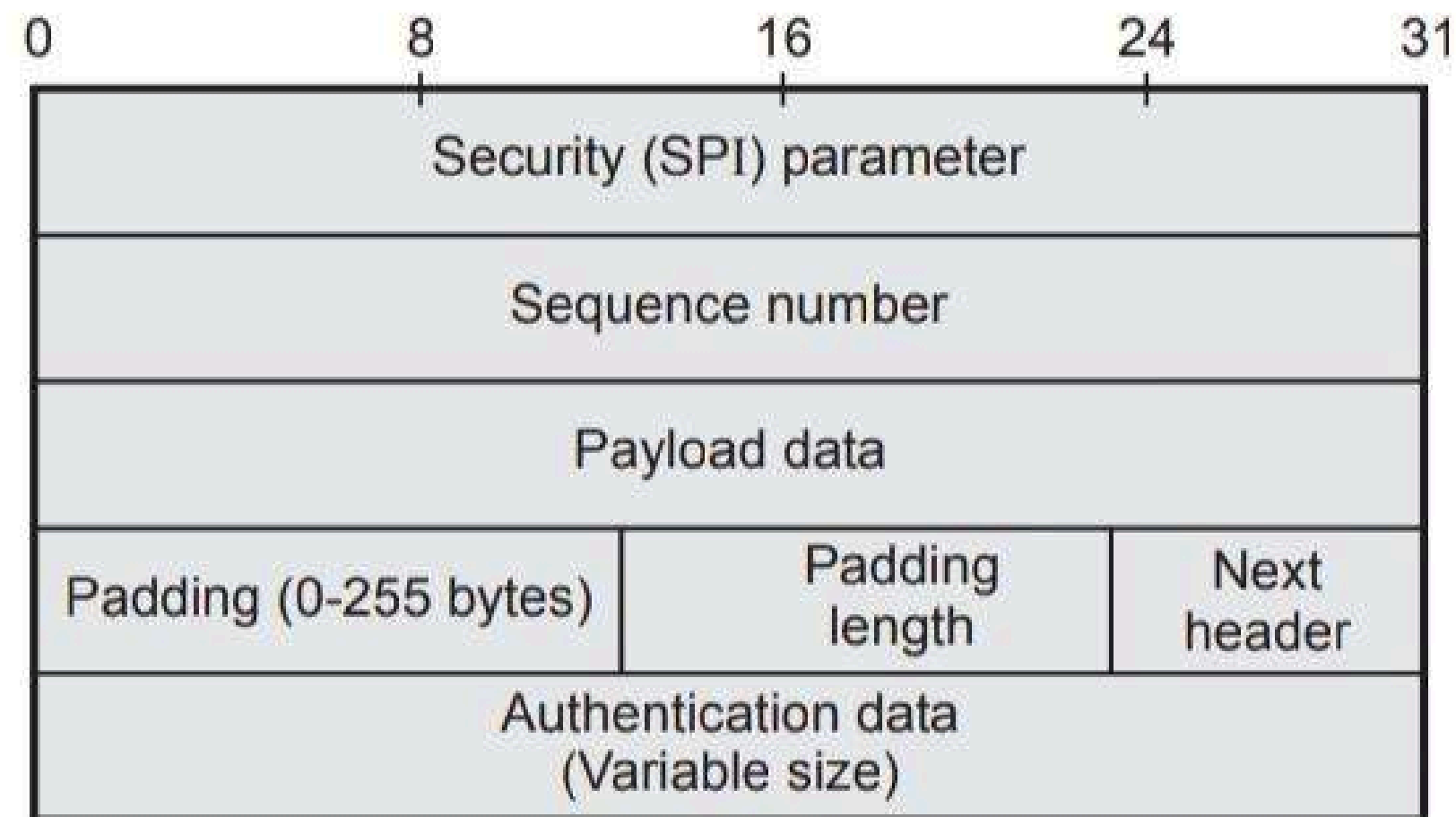


Fig. 4.15.1 ESP format

ESP Format

ESP Header Fields

1. Security Parameter Index (SPI)

- 32-bit field.
- Used with source and destination addresses.
- Identifies the Security Association (SA).

2. Sequence Number

- 32-bit field.
- Used to prevent Replay Attacks.
- Each packet is assigned a unique sequence number.

3. Payload Data

- Contains the actual data to be protected.

- In Transport Mode → contains transport layer segment.
- In Tunnel Mode → contains complete IP packet.
- Payload data is encrypted.

4. Padding

- Contains extra padding bits.
- Used for:
 - Proper data alignment
 - Hiding actual message length

5. Padding Length

ESP Format

- Indicates the number of padding bytes added before this field.

6. Next Header

- Identifies the type of data encapsulated in the payload.
- Specifies which protocol comes next.

7. Authentication Data

- Variable-length field.
- Contains authentication information called Integrity Check Value (ICV).
- Used for:
 - Data integrity and Authentication

Features of ESP

- Provides data confidentiality through encryption.
- Supports optional authentication.
- Protects against replay attacks.
- Works in both transport mode and tunnel mode.

Advantages of ESP

- Encrypts sensitive information.
- Provides secure communication over networks.
- Ensures confidentiality and integrity.



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